Cell Design Specification — T4R1FLASH

Case: `t4_r1_flash` | Generation date: 2026-08-25 | Doc: VBF-T4R1FLASH-DS-01

All values mechanical from simulation outputs / parameter set (see log.jsonl); no numbers from memory.

1. Basic specification

Field	Value	Source
Electrochemical system	NMC811 / graphite (Li-ion)	Chen2020 base parameter set (anchor-table mapping)
Nominal capacity	3.93 Ah (simulation-verified 1C: 3.9547 Ah)	parameter set + cell/final_b7_1c_dfn.json
Voltage window	2.5 – 4.2 V	parameter set (Upper/Lower voltage cut-off)
Nominal voltage (midpoint)	3.8142 V	cell/final_b7_energy_dfn.json
Electrode strip	65 mm height x 1580 mm width (wound strip)	parameter set (Electrode height/width)
Cell external dimensions	Not provided (no shell/can thickness parameter in set)	parameter set
Electrolyte	EC/EMC + LiPF6 base; transport overridden: sigma=2.0 S/m, D=5.0e-10 m2/s, t+=0.60	Chen2020 + design override (R2-R5)
Electrolyte concentration	1000 mol/m3 (bulk solvent 2636 mol/m3)	parameter set
Additive candidates (screened)	FEC / VC — molecular funnel: 2 passed, 0 rejected	cell/r2_funnel_*.json

2. Electrode and separator

Layer	Thickness (µm)	Porosity	Active vol. frac.	Density (kg/m3)	Source
Positive electrode (NMC811)	60.0	0.335	0.665	3262	parameter set (design override on thickness)
Negative electrode (graphite)	68.0	0.250	0.750	1657	parameter set (design override on thickness)
Separator	8.0	0.470	—	397	parameter set (design override on thickness)
Positive current collector (Al)	10.0	—	—	2700	parameter set (design override on thickness)
Negative current collector (Cu)	6.0	—	—	8960	parameter set (design override on thickness)
Particle radius	pos 5.22 µm / neg 5.86 µm	—	—	—	parameter set

N/P ratio = (c_max,neg x eps,neg x t,neg) / (c_max,pos x eps,pos x t,pos) = (33133 x 0.750 x 68.0) / (63104 x 0.665 x 60.0) = **0.671**

Design note: the protocol N/P formula uses maximum concentrations; the cell is positive-limited in operation (initial lithiation pos 0.27 / neg 0.90; 1C delivers 3.955 Ah ≥ nominal 3.93 Ah).

3. Process design parameters

Parameter	Value	Formula / note

Positive areal density	130.2 g/m2	thickness x (1-porosity) x density
Negative areal density	84.5 g/m2	thickness x (1-porosity) x density
Positive compaction density	2.17 g/cm3	density x (1-porosity), /1000
Negative compaction density	1.24 g/cm3	density x (1-porosity), /1000
Electrolyte fill amount	5.04 g/cell	pore volume x electrolyte density (1.2 g/cm3, lit.) x fill factor 1.0
Formation recommendation	0.1C CC to 4.2 V, 25 degC, 2 cycles	design recommended value; actual production-line value requires tuning

4. Mass breakdown (per cell)

Component	Mass (g)	Caliber
positive_electrode	13.37	contract formula: thickness x area x (1-porosity) x density
negative_electrode	8.68	contract formula: thickness x area x (1-porosity) x density
positive_cc	2.77	contract formula: thickness x area x (1-porosity) x density
negative_cc	5.52	contract formula: thickness x area x (1-porosity) x density
separator	0.17	contract formula: thickness x area x (1-porosity) x density
Total (electrolyte excluded)	**30.51**	= calc-energy mass_kg 30.51 g (cell/final_b7_energy_dfn.json)
Electrolyte (not in contract caliber)	5.04	pore volume x 1.2 g/cm3 (lit.)
Total incl. electrolyte	35.55	formula-caliber estimate

5. Performance verification (DFN, final candidate)

Metric	Value	Requirement (entry 0)	Verdict	Source
1C discharge capacity	3.955 Ah	nominal 3.93 Ah	✓	cell/final_b7_1c_dfn.json
-20 degC 1C retention	0.9929	≥ 0.95	✓	cell/final_b7_retention_dfn.json
Gravimetric energy density	460.1 Wh/kg	≥ 327.18	✓	cell/final_b7_energy_dfn.json
Volumetric energy density	899.3 Wh/L	≥ 880.0	✓	cell/final_b7_energy_dfn.json
4C/45 degC charge T_max	329.61 K	≤ 333.15 K	✓	cell/final_b7_4c45_dfn.json
4C plating (anode min)	+0.0193 V	> 0 V	✓	cell/final_b7_4c45_dfn.json
Overcharge -> thermal runaway	triggered = False	false	✓	cell/final_b7_tr.json

6. Design notes (what changed vs baseline and why)

1. **Electrode thickness 75/85 -> 60/68 µm; separator 25 -> 8 µm; CC 12/12 -> 10/6 µm** — volumetric energy density 843.5 -> 899.3 Wh/L (R2-R5; thickness levers had no effect on plating, measured).

2. **C_nom reset to design's true 1C capacity (3.93 Ah)** — C-rate consistency: 4C must mean the design's own 4C (R2 note).

3. **Electrolyte transport override (sigma 2.0 S/m, D 5e-10 m2/s, t+ 0.60)** — the ONLY lever that fixed 4C plating (anode min -0.44 V -> +0.019 V; thickness/particle levers measured ineffective, R2-R5).

4. **h = 40 W/m2K forced-air cooling** — 4C exam T_max 350.6 (baseline) -> 329.6 K; trend ~ -2 K per +5 W/m2K (R3-R5).

5. **-20 degC retention needs no insulation** — Chen2020 electrolyte/solid transport is temperature-independent (measured via parameter introspection); retention 99.3% maintained at all h (plan update entry).

6. **Thermal management tension resolved** — the planned insulation-vs-cooling trade-off does not materialize in this parameter set (same entry).